

MS&E 342: Stochastic Systems and Learning Theory with Applications in Finance

Lecture 1: Stochastic Calculus Review

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The goal of this lecture note is to provide a concise overview of stochastic calculus. The derivations are presented at an intuitive level, rather than in full mathematical rigor. We will focus on Sections 1-3 in class.

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1 Brownian motion and quadratic variation

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions.

Definition 1.1 (Brownian motion). A one-dimensional Brownian motion $(W_t)_{t \geq 0}$ is an adapted continuous process such that

1. $W_0 = 0$ almost surely,
2. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t$,
3. increments on disjoint intervals are independent.

The main reason ordinary calculus fails for W is that $(dW_t)^2$ is effectively of order dt , rather than a negligible higher-order term. Consequently, the second-order term survives in the stochastic expansion.

Proposition 1.2 (Quadratic variation of Brownian motion). *Let $\pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$ be a partition of $[0, t]$, and let*

$$Q_\pi := \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2.$$

Then $Q_\pi \rightarrow t$ in L^2 as $|\pi| \rightarrow 0$. Equivalently,

$$[W]_t = t.$$

Proof. Set $\Delta_i W := W_{t_{i+1}} - W_{t_i}$. Since $\Delta_i W \sim \mathcal{N}(0, \Delta_i t)$ and the increments are independent,

$$\mathbb{E} Q_\pi = \sum_i \mathbb{E}[(\Delta_i W)^2] = \sum_i \Delta_i t = t.$$

For a centered Gaussian $Z \sim \mathcal{N}(0, h)$,

$$\mathbb{E}[Z^4] = 3h^2, \quad \text{Var}(Z^2) = 3h^2 - h^2 = 2h^2.$$

Therefore,

$$\text{Var}(Q_\pi) = \sum_i \text{Var}((\Delta_i W)^2) = 2 \sum_i (\Delta_i t)^2 \leq 2|\pi| \sum_i \Delta_i t = 2t|\pi| \rightarrow 0.$$

Hence $Q_\pi \rightarrow t$ in L^2 . □

This is the origin of the symbolic rules

$$(dW_t)^2 = dt, \quad dt \, dW_t = 0, \quad (dt)^2 = 0.$$

They are not literal algebraic identities, but they are the correct bookkeeping rules inside Itô calculus.

Example 1.3. Applying the same computation to a differentiable path $x(t)$ gives

$$\sum_i (x(t_{i+1}) - x(t_i))^2 \rightarrow 0,$$

so smooth paths have zero quadratic variation. Brownian motion is therefore continuous but genuinely rough.

2 The Itô integral and martingales

Definition 2.1 (Martingale). An adapted process $M = (M_t)_{0 \leq t \leq T}$ is an $(\mathcal{F}_t)$ -martingale if

1. $M_t \in L^1$ for every $t \in [0, T]$,
2. for every $0 \leq s \leq t \leq T$,

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{a.s.}$$

Equivalently, the future increment has conditional mean zero:

$$\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = 0.$$

Remark 2.2. This is the probabilistic analogue of a “fair game”: once the present information $\mathcal{F}_s$ is fixed, the best prediction of the future value M_t is the current value M_s . The key point for us is that Itô integrals are martingales under square-integrability, so they behave like the drift-free part of a stochastic differential equation.

2.1 Definition on simple processes

For a simple adapted process

$$H_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t), \quad \xi_i \in L^2(\Omega, \mathcal{F}_{t_i}),$$

define

$$\int_0^t H_s \, dW_s := \sum_{i=0}^{n-1} \xi_i (W_{t \wedge t_{i+1}} - W_{t \wedge t_i}).$$

The point of adaptedness is that ξ_i is chosen using information available up to time t_i .

Theorem 2.3 (Itô isometry). *For every square-integrable simple adapted process H ,*

$$\mathbb{E} \left[\left(\int_0^t H_s \, dW_s \right)^2 \right] = \mathbb{E} \int_0^t H_s^2 \, ds.$$

Consequently the integral extends by completion to all adapted processes satisfying

$$\mathbb{E} \int_0^t H_s^2 \, ds < \infty.$$

Proof. Write $I_t(H) = \sum_i \xi_i \Delta_i W$ with $\Delta_i W = W_{t \wedge t_{i+1}} - W_{t \wedge t_i}$. Then

$$\mathbb{E}[I_t(H)^2] = \sum_i \mathbb{E}[\xi_i^2 (\Delta_i W)^2] + 2 \sum_{i < j} \mathbb{E}[\xi_i \xi_j \Delta_i W \Delta_j W].$$

For $i < j$, the variable $\xi_i \xi_j \Delta_i W$ is $\mathcal{F}_{t_j}$ -measurable, whereas $\Delta_j W$ is independent of $\mathcal{F}_{t_j}$ and centered, so

$$\mathbb{E}[\xi_i \xi_j \Delta_i W \Delta_j W] = 0.$$

Thus only the diagonal terms remain:

$$\mathbb{E}[I_t(H)^2] = \sum_i \mathbb{E}[\xi_i^2] (t \wedge t_{i+1} - t \wedge t_i) = \mathbb{E} \int_0^t H_s^2 \, ds.$$

□

Example 2.4 (Trading gains as an Itô integral). Let Δ_t be the number of shares held in a stock with price S_t . The gains process is

$$G_t = \int_0^t \Delta_s dS_s.$$

If

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

then

$$G_t = \int_0^t \Delta_s \mu S_s ds + \int_0^t \Delta_s \sigma S_s dW_s.$$

The second term is an Itô integral and represents the *random P&L*. In particular, if the strategy is self-financing and hedged, the stochastic term can be eliminated.

Proposition 2.5 (Martingale property). *If H is adapted and $\mathbb{E} \int_0^T H_s^2 ds < \infty$, then*

$$M_t := \int_0^t H_s dW_s$$

is an $(\mathcal{F}_t)$ -martingale and

$$[M]_t = \int_0^t H_s^2 ds.$$

Why the martingale property is natural. For a simple process,

$$M_t - M_s = \sum_{i: t_i > s} \xi_i (W_{t \wedge t_{i+1}} - W_{t \wedge t_i \vee s}).$$

Each increment after time s is independent of the past and has conditional mean zero given $\mathcal{F}_s$, so

$$\mathbb{E}[M_t - M_s \mid \mathcal{F}_s] = 0.$$

The bracket identity follows first for simple processes from quadratic variation, and then by L^2 -approximation. $\square$

Example 2.6 (Exponential martingale). For any $\lambda \in \mathbb{R}$,

$$\mathcal{E}_t(\lambda) := \exp\left(\lambda W_t - \frac{1}{2}\lambda^2 t\right)$$

is a martingale. Indeed, with $f(t, x) = \exp(\lambda x - \frac{1}{2}\lambda^2 t)$,

$$f_t = -\frac{1}{2}\lambda^2 f, \quad f_x = \lambda f, \quad f_{xx} = \lambda^2 f,$$

so Itô's formula gives

$$d\mathcal{E}_t(\lambda) = \lambda \mathcal{E}_t(\lambda) dW_t.$$

The drift vanishes, hence $\mathcal{E}_t(\lambda)$ is a local martingale; positivity plus unit expectation makes it a true martingale.

3 Itô's formula, generators, and PDEs

3.1 Itô's formula

Let

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t.$$

Theorem 3.1 (Itô's formula in one dimension). *If $f \in C^{1,2}([0, T] \times \mathbb{R})$, then*

$$df(t, X_t) = \left(f_t + bf_x + \frac{1}{2}\sigma^2 f_{xx} \right)(t, X_t) dt + f_x(t, X_t)\sigma(t, X_t) dW_t.$$

Heuristic derivation. Taylor expansion up to second order gives

$$\Delta f \approx f_t \Delta t + f_x \Delta X + \frac{1}{2} f_{xx} (\Delta X)^2.$$

Now

$$\Delta X = b\Delta t + \sigma\Delta W,$$

so

$$(\Delta X)^2 = b^2(\Delta t)^2 + 2b\sigma\Delta t\Delta W + \sigma^2(\Delta W)^2.$$

The first two terms are negligible at order Δt , whereas $(\Delta W)^2 \sim \Delta t$. Thus

$$(\Delta X)^2 \sim \sigma^2 \Delta t,$$

which yields

$$\Delta f \approx f_t \Delta t + f_x \Delta X + \frac{1}{2} f_{xx} \sigma^2 \Delta t.$$

This is exactly the formula above. □

Example 3.2 (The prototype martingale). Taking $X_t = W_t$ and $f(x) = x^2$, we get

$$d(W_t^2) = 2W_t dW_t + dt.$$

Hence

$$W_t^2 - t$$

is a martingale.

Remark 3.3. This example shows how Itô's correction term creates a compensator. For a smooth path $x(\cdot)$, one has

$$d(x_t^2) = 2x_t dx_t.$$

For Brownian motion, however,

$$d(W_t^2) = 2W_t dW_t + dt,$$

because $(dW_t)^2 = dt$. The extra dt is exactly the finite-variation term that compensates for the drift of W_t^2 . Indeed,

$$d(W_t^2 - t) = 2W_t dW_t,$$

so $W_t^2 - t$ is a martingale, whereas W_t^2 itself is not.

Example 3.4 (Logarithm of geometric Brownian motion). Suppose

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_t > 0.$$

Applying Itô's formula to $f(x) = \log x$ gives

$$d \log S_t = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (dS_t)^2 = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t.$$

Therefore

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right).$$

3.2 Generator and Dynkin's formula

For the diffusion above define the generator

$$(L_t f)(x) := b(t, x) f'(x) + \frac{1}{2} \sigma(t, x)^2 f''(x).$$

In d dimensions, if

$$dX_t = b(t, X_t) dt + \Sigma(t, X_t) dW_t, \quad a = \Sigma \Sigma^\top,$$

then

$$(L_t f)(x) = b(t, x) \cdot \nabla f(x) + \frac{1}{2} \text{Tr}(a(t, x) \nabla^2 f(x)).$$

Proposition 3.5 (Local martingale generated by Itô). *If $f \in C^{1,2}$, then*

$$M_t^f := f(t, X_t) - f(0, X_0) - \int_0^t (\partial_s + L_s) f(s, X_s) ds$$

is a local martingale.

Proof. This is just Itô's formula rewritten so that the drift is moved to the left-hand side:

$$M_t^f = \int_0^t f'(t, X_s) \sigma(t, X_s) dW_s.$$

□

Corollary 3.6 (Dynkin's formula). *If integrability is sufficient to take expectations, then*

$$\mathbb{E}[f(t, X_t)] = f(0, X_0) + \mathbb{E} \int_0^t (\partial_s + L_s) f(s, X_s) ds.$$

More generally, if $X^{t,x}$ denotes the solution started from $X_t^{t,x} = x$, then for $0 \leq t \leq \tau \leq T$,

$$\mathbb{E}[f(\tau, X_\tau^{t,x})] = f(t, x) + \mathbb{E} \int_t^\tau (\partial_s + L_s) f(s, X_s^{t,x}) ds.$$

Dynkin's formula provides the standard link between stochastic differential equations and linear parabolic PDEs.

Remark 3.7 (Interpretation of Dynkin's formula). Dynkin's formula is the expectation-level version of Itô's formula. The local martingale term disappears after taking expectations, so the expected evolution of $f(t, X_t)$ is governed entirely by the operator $\partial_t + L_t$:

$$\mathbb{E}[f(t, X_t)] = f(0, X_0) + \mathbb{E} \int_0^t (\partial_s + L_s) f(s, X_s) ds.$$

Thus L_t should be viewed as the infinitesimal mean rate of change of $f(X_t)$. In particular, if u satisfies

$$(\partial_t + L_t)u = 0,$$

then $u(t, X_t)$ has no drift, which is the probabilistic origin of the backward Kolmogorov equation and the martingale method for PDEs.

Proposition 3.8 (Kolmogorov backward equation). *For each $(t, x) \in [0, T] \times \mathbb{R}$, let $X^{t,x}$ denote the solution of*

$$dX_s^{t,x} = b(s, X_s^{t,x}) ds + \sigma(s, X_s^{t,x}) dW_s, \quad s \in [t, T], \quad X_t^{t,x} = x.$$

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be bounded and smooth, and define

$$u(t, x) := \mathbb{E}[\phi(X_T) \mid X_t = x].$$

Assume that $u \in C^{1,2}([0, T] \times \mathbb{R}) \cap C([0, T] \times \mathbb{R})$. Then

$$\partial_t u(t, x) + (L_t u)(t, x) = 0, \quad 0 \leq t < T,$$

with terminal condition

$$u(T, x) = \phi(x).$$

The same statement holds in d dimensions with the generator defined above.

Formal derivation. Fix $0 \leq t < T$ and $h > 0$ with $t + h \leq T$. By the Markov property,

$$u(t, x) = \mathbb{E}[u(t + h, X_{t+h}^{t,x})].$$

Applying Itô's formula to the process $s \mapsto u(s, X_s^{t,x})$ on $[t, t + h]$ gives

$$u(t + h, X_{t+h}^{t,x}) - u(t, x) = \int_t^{t+h} (\partial_s + L_s) u(s, X_s^{t,x}) ds + \int_t^{t+h} u_x(s, X_s^{t,x}) \sigma(s, X_s^{t,x}) dW_s.$$

Taking expectations, the stochastic integral has mean zero, so

$$0 = \mathbb{E} \int_t^{t+h} (\partial_s + L_s) u(s, X_s^{t,x}) ds.$$

Divide by h and send $h \downarrow 0$ to obtain

$$\partial_t u(t, x) + (L_t u)(t, x) = 0.$$

The terminal condition is immediate from $X_T^{T,x} = x$:

$$u(T, x) = \mathbb{E}[\phi(X_T^{T,x})] = \phi(x).$$

□

Remark 3.9 (Connection to Feynman–Kac formula.). The PDE

$$\partial_t u + L_t u = 0, \quad u(T, x) = \phi(x),$$

is the backward Kolmogorov equation associated with the diffusion X . The identity

$$u(t, x) = \mathbb{E}[\phi(X_T^{t,x})]$$

is its probabilistic representation, and may be viewed as the simplest form of the Feynman–Kac formula (corresponding to zero potential and no source term).

Proposition 3.10 (Fokker–Planck equation). *Assume that X_t admits a smooth density $p(t, \cdot)$ on $\mathbb{R}^d$. Then*

$$\partial_t p(t, x) = -\nabla \cdot (b(t, x)p(t, x)) + \frac{1}{2} \sum_{i,j=1}^d \partial_{ij} (a_{ij}(t, x)p(t, x)),$$

where $a(t, x) = \Sigma(t, x)\Sigma(t, x)^\top$.

Proof sketch via duality argument. For any smooth compactly supported test function φ ,

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x)p(t, x) \, dx = \frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \mathbb{E}[(L_t \varphi)(X_t)] = \int_{\mathbb{R}^d} (L_t \varphi)(x)p(t, x) \, dx.$$

Now integrate by parts to move derivatives from φ onto p . The adjoint operator is

$$L_t^* p = -\nabla \cdot (bp) + \frac{1}{2} \sum_{i,j} \partial_{ij} (a_{ij}p),$$

so $\partial_t p = L_t^* p$.

Detailed proof. Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ be arbitrary. Since $p(t, \cdot)$ is the density of X_t ,

$$\int_{\mathbb{R}^d} \varphi(x)p(t, x) \, dx = \mathbb{E}[\varphi(X_t)].$$

Differentiating in t , we obtain

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x)p(t, x) \, dx = \frac{d}{dt} \mathbb{E}[\varphi(X_t)].$$

By Dynkin’s formula applied to the time-independent test function φ , we have

$$\mathbb{E}[\varphi(X_t)] = \mathbb{E}[\varphi(X_0)] + \mathbb{E} \int_0^t (L_s \varphi)(X_s) \, ds,$$

hence, under the stated smoothness assumptions,

$$\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \mathbb{E}[(L_t \varphi)(X_t)].$$

Using again that $p(t, \cdot)$ is the density of X_t , this gives

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x)p(t, x) \, dx = \int_{\mathbb{R}^d} (L_t \varphi)(x)p(t, x) \, dx.$$

Now write the generator in coordinates:

$$(L_t\varphi)(x) = \sum_{i=1}^d b_i(t, x) \partial_i \varphi(x) + \frac{1}{2} \sum_{i,j=1}^d a_{ij}(t, x) \partial_{ij} \varphi(x).$$

Therefore

$$\int_{\mathbb{R}^d} (L_t\varphi)(x) p(t, x) \, dx = \sum_{i=1}^d \int_{\mathbb{R}^d} b_i(t, x) \partial_i \varphi(x) p(t, x) \, dx + \frac{1}{2} \sum_{i,j=1}^d \int_{\mathbb{R}^d} a_{ij}(t, x) \partial_{ij} \varphi(x) p(t, x) \, dx.$$

We integrate by parts term by term. Since φ has compact support, there are no boundary terms. For the first-order part,

$$\int_{\mathbb{R}^d} b_i(t, x) \partial_i \varphi(x) p(t, x) \, dx = - \int_{\mathbb{R}^d} \varphi(x) \partial_i (b_i(t, x) p(t, x)) \, dx.$$

Summing over i yields

$$\sum_{i=1}^d \int_{\mathbb{R}^d} b_i \partial_i \varphi p \, dx = - \int_{\mathbb{R}^d} \varphi(x) \nabla \cdot (b(t, x) p(t, x)) \, dx.$$

For the second-order part, integrate by parts twice:

$$\int_{\mathbb{R}^d} a_{ij}(t, x) \partial_{ij} \varphi(x) p(t, x) \, dx = \int_{\mathbb{R}^d} \varphi(x) \partial_{ij} (a_{ij}(t, x) p(t, x)) \, dx.$$

Hence

$$\frac{1}{2} \sum_{i,j=1}^d \int_{\mathbb{R}^d} a_{ij}(t, x) \partial_{ij} \varphi(x) p(t, x) \, dx = \frac{1}{2} \sum_{i,j=1}^d \int_{\mathbb{R}^d} \varphi(x) \partial_{ij} (a_{ij}(t, x) p(t, x)) \, dx.$$

Combining the two parts, we obtain

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) p(t, x) \, dx = \int_{\mathbb{R}^d} \varphi(x) \left[-\nabla \cdot (b(t, x) p(t, x)) + \frac{1}{2} \sum_{i,j=1}^d \partial_{ij} (a_{ij}(t, x) p(t, x)) \right] \, dx.$$

On the other hand,

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi(x) p(t, x) \, dx = \int_{\mathbb{R}^d} \varphi(x) \partial_t p(t, x) \, dx.$$

Therefore

$$\int_{\mathbb{R}^d} \varphi(x) \left[\partial_t p(t, x) + \nabla \cdot (b(t, x) p(t, x)) - \frac{1}{2} \sum_{i,j=1}^d \partial_{ij} (a_{ij}(t, x) p(t, x)) \right] \, dx = 0$$

for every $\varphi \in C_c^\infty(\mathbb{R}^d)$. Since the test function φ is arbitrary, it follows that

$$\partial_t p(t, x) = -\nabla \cdot (b(t, x) p(t, x)) + \frac{1}{2} \sum_{i,j=1}^d \partial_{ij} (a_{ij}(t, x) p(t, x)),$$

which is the Fokker–Planck equation. □

This equation is worth emphasizing, since generators and their adjoints provide the basic analytic description of how a diffusion evolves both at the path level and at the density level.

4 Application as Itô calculus: Black–Scholes

We model the stock price by geometric Brownian motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad dB_t = r B_t dt,$$

where $B_t = e^{rt}$ is the bank account.

Let $V(t, S)$ denote the price of a European claim with payoff $\Phi(S_T)$ at maturity T .

4.1 Derivation of the PDE

By Itô's formula,

$$dV(t, S_t) = \left(V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt + \sigma S_t V_S dW_t.$$

Consider the self-financing portfolio

$$\Pi_t := \Delta_t S_t - V(t, S_t).$$

Then

$$d\Pi_t = \Delta_t dS_t - dV(t, S_t).$$

Choose the hedge ratio

$$\Delta_t = V_S(t, S_t).$$

The stochastic term cancels, leaving

$$d\Pi_t = - \left(V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS} \right) dt.$$

This portfolio is locally riskless, so in an arbitrage-free market it must earn the risk-free rate:

$$d\Pi_t = r \Pi_t dt = r(S_t V_S - V) dt.$$

Comparing the two drift terms yields the Black–Scholes PDE:

$$V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + r S V_S - r V = 0, \quad V(T, S) = \Phi(S).$$

The striking point is that the drift μ disappears. The price depends on replication and no-arbitrage, not on the stock's physical expected return.

4.2 Martingale form of the pricing relation

Assume now directly that under a risk-neutral measure $\mathbb{Q}$,

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}}.$$

Applying Itô's formula to the discounted price process gives

$$d(e^{-rt} V(t, S_t)) = e^{-rt} \left(V_t + r S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS} - r V \right) dt + e^{-rt} \sigma S V_S dW_t^{\mathbb{Q}}.$$

Hence the Black–Scholes PDE is exactly the condition that

$$(e^{-rt}V(t, S_t))_{t \leq T}$$

be a martingale. Therefore

$$V(t, S_t) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}\Phi(S_T) \mid \mathcal{F}_t].$$

So the same Itô calculation simultaneously produces

- the hedging PDE,
- the martingale pricing identity,
- the interpretation of discounting as drift removal.

4.3 Closed form for the European call

For the call payoff $\Phi(s) = (s - K)^+$, under $\mathbb{Q}$ we have

$$\log S_T \mid \mathcal{F}_t \sim \mathcal{N}\left(\log S_t + \left(r - \frac{1}{2}\sigma^2\right)\tau, \sigma^2\tau\right), \quad \tau := T - t.$$

A standard lognormal computation gives

$$C(t, S_t) = S_t N(d_1) - K e^{-r\tau} N(d_2),$$

where

$$d_1 = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

5 Gaussian smoothing and Tweedie's formula

5.1 The static Gaussian perturbation identity

Let X be an $\mathbb{R}^d$ -valued random vector with density p_0 , let $Z \sim \mathcal{N}(0, I_d)$ be independent of X , and define

$$Y = X + \sigma Z.$$

Then the density of Y is the Gaussian smoothing

$$p_\sigma(y) = \int_{\mathbb{R}^d} p_0(x) \varphi_\sigma(y - x) \, dx,$$

where

$$\varphi_\sigma(z) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{|z|^2}{2\sigma^2}\right).$$

Proposition 5.1 (Tweedie's formula). *For every $y \in \mathbb{R}^d$ such that $p_\sigma(y) > 0$,*

$$\mathbb{E}[X \mid Y = y] = y + \sigma^2 \nabla \log p_\sigma(y).$$

Equivalently,

$$\nabla \log p_\sigma(y) = \frac{1}{\sigma^2} (\mathbb{E}[X \mid Y = y] - y).$$

Proof. Differentiate the smoothed density under the integral sign:

$$\nabla p_\sigma(y) = \int p_0(x) \nabla \varphi_\sigma(y - x) \, dx.$$

Since

$$\nabla \varphi_\sigma(y - x) = -\frac{y - x}{\sigma^2} \varphi_\sigma(y - x),$$

we obtain

$$\nabla p_\sigma(y) = -\frac{1}{\sigma^2} \int (y - x) p_0(x) \varphi_\sigma(y - x) \, dx = \frac{1}{\sigma^2} \int (x - y) p_0(x) \varphi_\sigma(y - x) \, dx.$$

Divide by $p_\sigma(y)$:

$$\nabla \log p_\sigma(y) = \frac{1}{\sigma^2} \frac{\int (x - y) p_0(x) \varphi_\sigma(y - x) \, dx}{\int p_0(x) \varphi_\sigma(y - x) \, dx}.$$

The fraction is exactly the conditional expectation $\mathbb{E}[X - y \mid Y = y]$ by Bayes' rule. Rearranging gives the result. $\square$

Remark 5.2 (Why this matters). The least-squares estimator of X from Y is the conditional mean $m^*(y) = \mathbb{E}[X \mid Y = y]$. Tweedie's formula says that for Gaussian corruption,

$$m^*(y) - y = \sigma^2 \nabla \log p_\sigma(y).$$

So a denoiser and a score field contain the same information, up to a simple affine transformation.

5.2 A more general version of Gaussian perturbation

Often one considers a more general Gaussian perturbation of the form

$$Y_t = \mu_t(X_0) + \sigma_t Z, \quad Z \sim \mathcal{N}(0, I_d),$$

where $\mu_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a deterministic (typically smooth) map and $\sigma_t > 0$.

Denote by p_t the density of Y_t . Then

$$p_t(y) = \int_{\mathbb{R}^d} p_0(x) \varphi_{\sigma_t}(y - \mu_t(x)) \, dx.$$

Proposition 5.3 (Generalized Tweedie identity). *Assume μ_t is continuously differentiable and invertible. Then for all y with $p_t(y) > 0$,*

$$\mathbb{E}[\mu_t(X_0) \mid Y_t = y] = y + \sigma_t^2 \nabla \log p_t(y).$$

Equivalently,

$$\nabla \log p_t(y) = \frac{1}{\sigma_t^2} (\mathbb{E}[\mu_t(X_0) \mid Y_t = y] - y).$$

Proof. Differentiate p_t under the integral:

$$\nabla p_t(y) = \int p_0(x) \nabla \varphi_{\sigma_t}(y - \mu_t(x)) \, dx.$$

Using

$$\nabla \varphi_{\sigma_t}(y - \mu_t(x)) = -\frac{y - \mu_t(x)}{\sigma_t^2} \varphi_{\sigma_t}(y - \mu_t(x)),$$

we obtain

$$\nabla p_t(y) = \frac{1}{\sigma_t^2} \int (\mu_t(x) - y) p_0(x) \varphi_{\sigma_t}(y - \mu_t(x)) \, dx.$$

Dividing by $p_t(y)$ gives

$$\nabla \log p_t(y) = \frac{1}{\sigma_t^2} \frac{\int (\mu_t(x) - y) p_0(x) \varphi_{\sigma_t}(y - \mu_t(x)) \, dx}{\int p_0(x) \varphi_{\sigma_t}(y - \mu_t(x)) \, dx}.$$

The ratio is $\mathbb{E}[\mu_t(X_0) - y \mid Y_t = y]$, which yields the result. $\square$

Remark 5.4. Two important special cases are:

- (Additive noise) $\mu_t(x) = x$, recovering the classical Tweedie formula.
- (Linear contraction) $\mu_t(x) = \alpha_t x$, giving

$$\mathbb{E}[X_0 \mid Y_t = y] = \frac{1}{\alpha_t} \left(y + \sigma_t^2 \nabla \log p_t(y) \right).$$

5.3 Time reversal

Consider the Itô SDE

$$dX_t = f(t, X_t) \, dt + g(t) \, dW_t,$$

where $g(t) > 0$ is scalar and let $p(t, \cdot)$ denote the density of X_t .

The density evolves according to the Fokker–Planck equation

$$\partial_t p(t, x) = -\nabla \cdot (f(t, x) p(t, x)) + \frac{1}{2} g(t)^2 \Delta p(t, x).$$

Proposition 5.5 (Time-reversal of diffusions). *Assume $p(t, x)$ is smooth and strictly positive. Define the time-reversed process*

$$\tilde{X}_s := X_{T-s}, \quad 0 \leq s \leq T.$$

Then $\tilde{X}$ satisfies the SDE

$$d\tilde{X}_s = \left(-f(T-s, \tilde{X}_s) + g(T-s)^2 \nabla \log p(T-s, \tilde{X}_s) \right) ds + g(T-s) \, d\tilde{W}_s,$$

where $\tilde{W}$ is a Brownian motion in the reversed filtration.

Remark 5.6 (Connection with Gaussian perturbation). The additional drift term

$$g(t)^2 \nabla \log p(t, x)$$

is the same score function that appears in Tweedie’s formula.

Heuristically:

- Tweedie’s formula identifies the score from a *single Gaussian perturbation*.

- Time reversal shows that the same score controls the *dynamics* of reversing a diffusion.

In particular, in the pure noise case $f \equiv 0$, the forward process smooths the distribution via Gaussian convolution, while the reverse process uses $\nabla \log p$ to reconstruct structure.

Remark 5.7 (Generator perspective). Let L_t be the generator of the forward process:

$$L_t \varphi = f(t, x) \cdot \nabla \varphi + \frac{1}{2} g(t)^2 \Delta \varphi.$$

Then the reversed process has generator

$$\tilde{L}_t \varphi = \left(-f(t, x) + g(t)^2 \nabla \log p(t, x) \right) \cdot \nabla \varphi + \frac{1}{2} g(t)^2 \Delta \varphi.$$

References

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